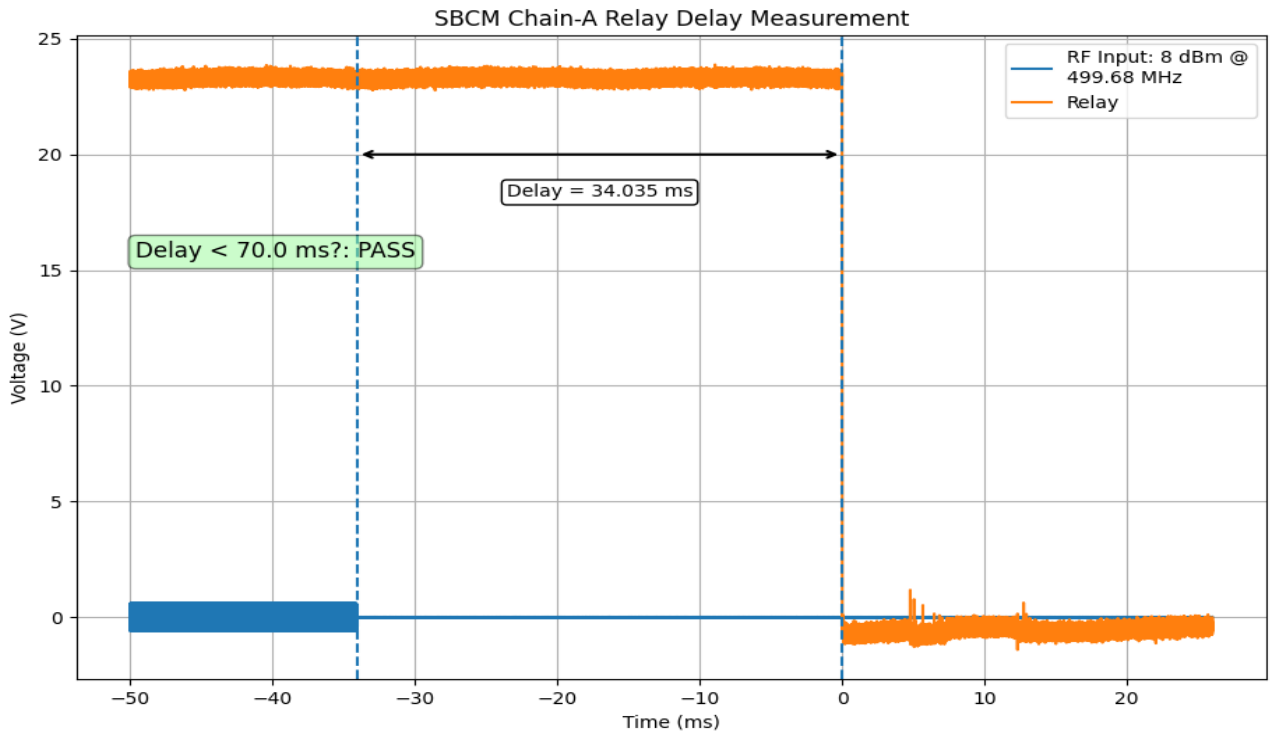
Results for SBCM Calibration for TOSS: NSLSII-INS-PRC-005
Procedure Performed on 1/17/2026 by Kato Bouthsarath and Anthony Caracappa
Analysis Performed on 1/21/2026 by Kato Bouthsarath

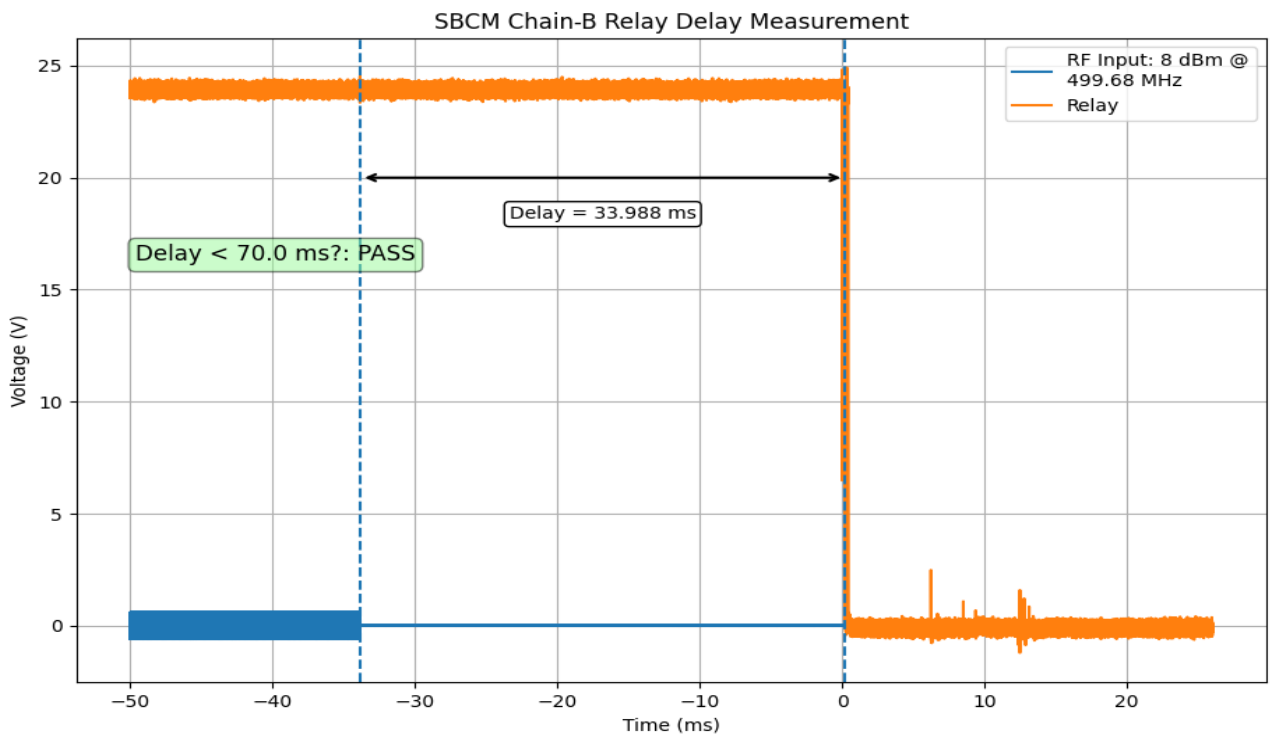
Test Equipment Used for Calibration		
Oscilloscope	Manufacturer	Tektronix
	Model Number	MSO64B
	Serial Number	B013046
	Calibration Expiration	12/29/2026
RF Source	Manufacturer	Rohde & Schwarz
	Model Number	SMB100B
	Serial Number	102463
	Calibration Expiration	5/8/2026
DMM1	Manufacturer	Keithley
	Model Number	2100
	Serial Number	8020356
	Calibration Expiration	12/29/2026
DMM2	Manufacturer	Keithley
	Model Number	2100
	Serial Number	8020357
	Calibration Expiration	12/29/2026
Python Scripts	Bandwidth Scan	BandwidthScan.py
	Characteristic Curve	CharacteristicCurve.py
	Beam Scan	BeamScan.py
	Report Generation	FunctionalTest.py
	DCCT Report	ProcessSBCM.py

Voltage Supply Measurements		
Actual Voltage (V)	Measured Voltage (V)	Status
-12.0 V	-12.025	PASS
+12.0 V	+12.004	PASS
-5.0 V	-5.081	PASS
+5.0 V	+5.118	PASS

Preliminary Settings Adjustments		
Adjustments made on 1/17/2026 by the CE: Kato Bouthsarath		
Adjustment	Set To:	Status
Rectifier Offset A	5.263mV	PASS
Rectifier Offset B	34.365mV	PASS
Downconverted Frequency A	1.773MHz	PASS
Downconverted Frequency B	1.804MHz	PASS
Measurement	Reading	Status
Downconverted Frequency A Level	266.2mVpp	PASS
Downconverted Frequency B Level	247.8mVpp	PASS

SBCM Relay Delay Measurements		
The relay delay is measured as the time difference between the falling edge of the RF input signal and the falling edge of the relay output signal.		
	Measured Delay (ms)	Status
Chain A Relay Delay	34.035	PASS
Chain B Relay Delay	33.988	PASS





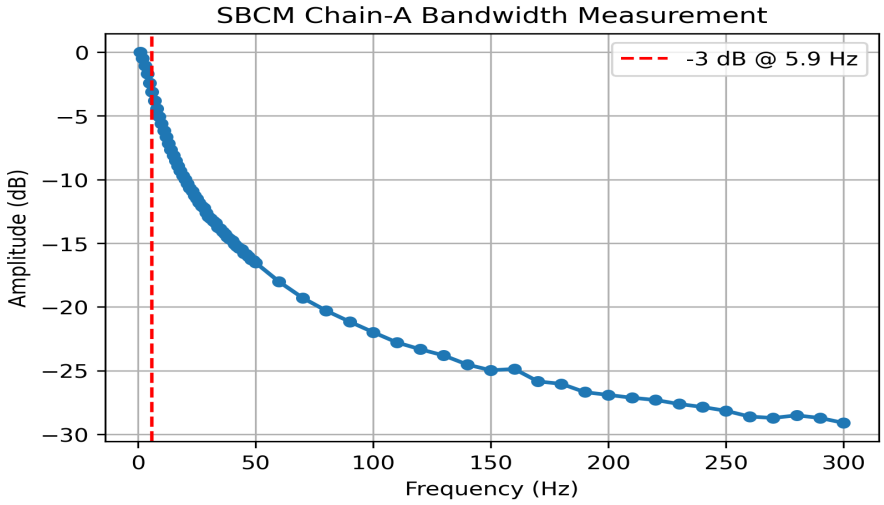
SBCM Chain-A Bandwidth Measurement				file: SBCM-A_2026_BWScan.txt				
Freq (Hz)	Vpp (V)	Amp (dB)	Freq (Hz)	Vpp (V)	Amp (dB)	Freq (Hz)	Vpp (V)	Amp (dB)
1	4.6608	0.0	26	1.1967	-11.81	60	0.5852	-18.02
2	4.4188	-0.46	27	1.1608	-12.07	70	0.5058	-19.29
3	4.1261	-1.06	28	1.1383	-12.24	80	0.4505	-20.3
4	3.8331	-1.7	29	1.0898	-12.62	90	0.4078	-21.16
5	3.5314	-2.41	30	1.0547	-12.91	100	0.3708	-21.99
6	3.2612	-3.1	31	1.0383	-13.04	110	0.3381	-22.79
7	3.0086	-3.8	32	1.0109	-13.27	120	0.3178	-23.33
8	2.797	-4.44	33	0.997	-13.4	130	0.3005	-23.81
9	2.6091	-5.04	34	0.9567	-13.75	140	0.2767	-24.53
10	2.4456	-5.6	35	0.9453	-13.86	150	0.2628	-24.98
11	2.2939	-6.16	36	0.9177	-14.12	160	0.2659	-24.87
12	2.1641	-6.66	37	0.8997	-14.29	170	0.2377	-25.85
13	2.0455	-7.15	38	0.8781	-14.5	180	0.2325	-26.04
14	1.9283	-7.67	39	0.8619	-14.66	190	0.2156	-26.7
15	1.8336	-8.1	40	0.8491	-14.79	200	0.2103	-26.91
16	1.7509	-8.5	41	0.8236	-15.05	210	0.205	-27.13
17	1.6673	-8.93	42	0.8056	-15.25	220	0.2009	-27.31
18	1.5956	-9.31	43	0.7941	-15.37	230	0.1938	-27.62
19	1.5275	-9.69	44	0.7839	-15.48	240	0.1886	-27.86
20	1.4798	-9.96	45	0.7556	-15.8	250	0.1822	-28.16
21	1.418	-10.34	46	0.7508	-15.86	260	0.173	-28.61
22	1.367	-10.65	47	0.7364	-16.03	270	0.1708	-28.72
23	1.3281	-10.9	48	0.7161	-16.27	280	0.1748	-28.52
24	1.2747	-11.26	49	0.7088	-16.36	290	0.1708	-28.72
25	1.2388	-11.51	50	0.6953	-16.53	300	0.1633	-29.11

The RF input is set to 499.68 MHz @ 0 dBm.

Freq: Amplitude Modulation Hz between 1 Hz and 300 Hz.

Vpp: Measured voltage (pk-pk) of the demodulated signal coming from the Recitifier Module.

Amp: Measured amplitude (dB) of the demodulated signal coming from the Recitifer Module.
0 dB is referring to the first point of the data.



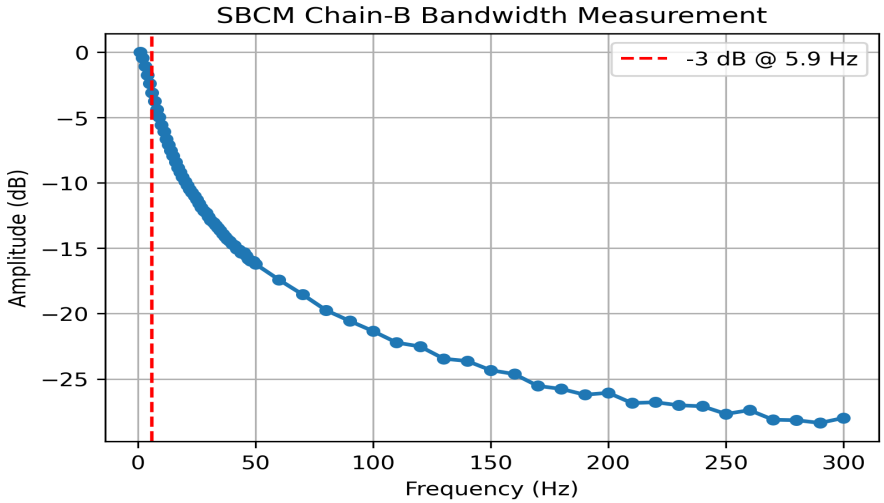
SBCM Chain-B Bandwidth Measurement				file: SBCM-B_2026_BWScan.txt				
Freq (Hz)	Vpkpk (V)	Amp (dB)	Freq (Hz)	Vpkpk (V)	Amp (dB)	Freq (Hz)	Vpkpk (V)	Amp (dB)
1	4.4291	0.0	26	1.1692	-11.57	60	0.5966	-17.41
2	4.2173	-0.43	27	1.13	-11.86	70	0.5242	-18.54
3	3.9208	-1.06	28	1.0952	-12.14	80	0.4561	-19.75
4	3.6214	-1.75	29	1.078	-12.27	90	0.4153	-20.56
5	3.358	-2.4	30	1.038	-12.6	100	0.3794	-21.34
6	3.1006	-3.1	31	1.0081	-12.86	110	0.343	-22.22
7	2.8755	-3.75	32	0.9872	-13.04	120	0.3316	-22.51
8	2.6744	-4.38	33	0.9659	-13.23	130	0.2977	-23.45
9	2.5042	-4.95	34	0.9423	-13.44	140	0.2917	-23.63
10	2.3294	-5.58	35	0.9219	-13.63	150	0.2689	-24.33
11	2.1975	-6.09	36	0.8942	-13.9	160	0.2602	-24.62
12	2.0631	-6.64	37	0.8723	-14.11	170	0.2342	-25.53
13	1.9559	-7.1	38	0.8533	-14.3	180	0.2281	-25.76
14	1.8594	-7.54	39	0.8384	-14.46	190	0.2167	-26.21
15	1.7775	-7.93	40	0.8181	-14.67	200	0.2205	-26.06
16	1.6838	-8.4	41	0.8081	-14.78	210	0.2014	-26.84
17	1.6019	-8.83	42	0.7827	-15.05	220	0.2028	-26.78
18	1.5398	-9.18	43	0.7755	-15.13	230	0.1975	-27.01
19	1.4725	-9.57	44	0.7569	-15.35	240	0.1958	-27.09
20	1.4198	-9.88	45	0.7588	-15.32	250	0.183	-27.68
21	1.372	-10.18	46	0.7395	-15.55	260	0.1892	-27.39
22	1.3214	-10.51	47	0.7158	-15.83	270	0.1739	-28.12
23	1.2891	-10.72	48	0.705	-15.96	280	0.1731	-28.16
24	1.2469	-11.01	49	0.7012	-16.01	290	0.1691	-28.37
25	1.2114	-11.26	50	0.6855	-16.21	300	0.1767	-27.98

The RF input is set to 499.68 MHz @ 0 dBm.

Freq: Amplitude Modulation Hz between 1 Hz and 300 Hz.

Vpkpk: Measured voltage (pk-pk) of the demodulated signal coming from the Recitifier Module.

Amp: Measured amplitude (dB) of the demodulated signal coming from the Recitifer Module.
0 dB is referring to the first point of the data.

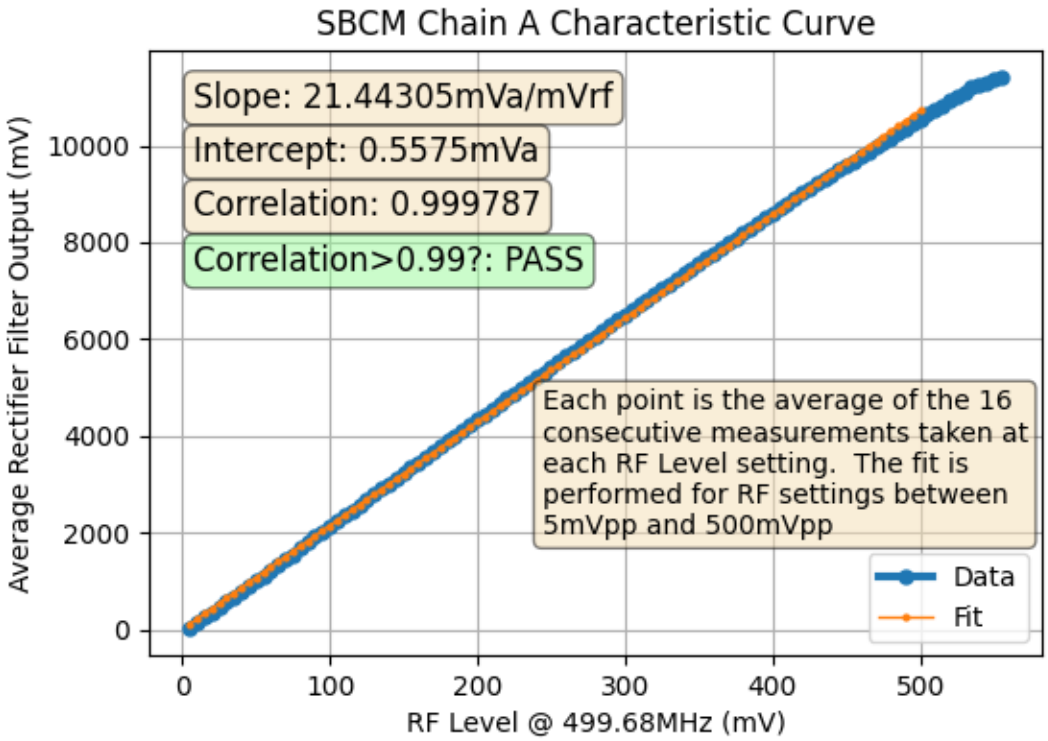


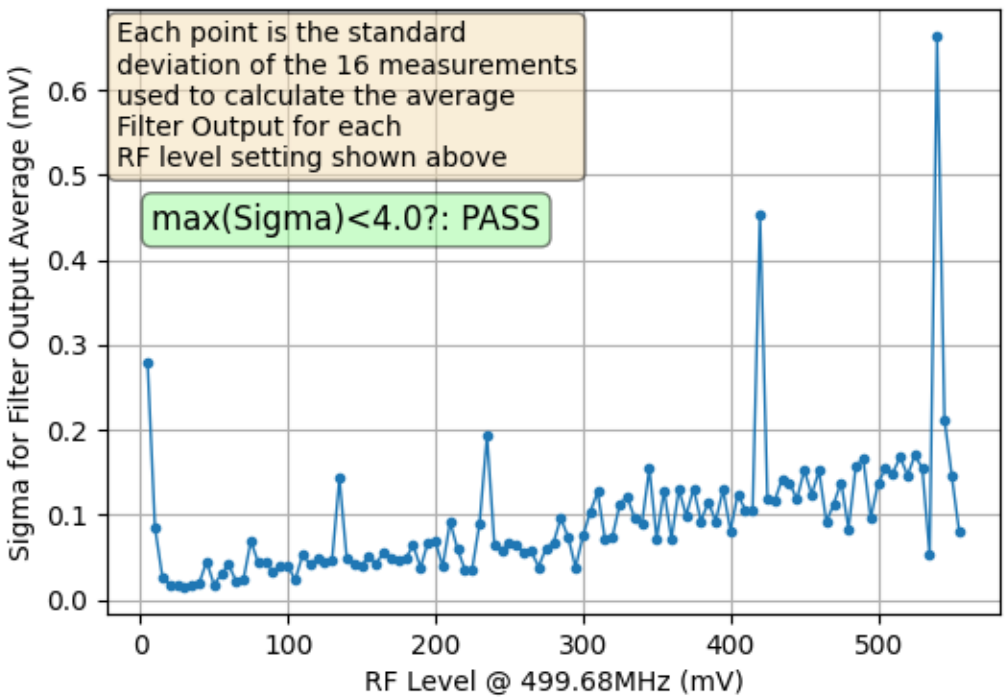
SBCM Chain-A Characteristic Curve				file: SBCM-A_2026_CharCurve.txt				
SetPnt (mVpp)	Filter (mV)	Sigma (mV)	SetPnt (mVpp)	Filter (mV)	Sigma (mV)	SetPnt (mVpp)	Filter (mV)	Sigma (mV)
5	36.28	0.2792	190	4113.77	0.0377	375	8084.44	0.1309
10	137.81	0.0849	195	4222.91	0.0676	380	8189.42	0.0925
15	243.67	0.0272	200	4332.57	0.0682	385	8292.37	0.1148
20	350.63	0.017	205	4442.01	0.039	390	8395.61	0.0925
25	459.42	0.0177	210	4551.33	0.0917	395	8500.52	0.1298
30	567.73	0.0155	215	4660.54	0.0598	400	8603.35	0.0813
35	676.15	0.0161	220	4769.9	0.0353	405	8705.13	0.1243
40	784.85	0.0192	225	4878.51	0.0362	410	8809.74	0.1053
45	895.1	0.0451	230	4987.73	0.0901	415	8912.12	0.1047
50	1004.14	0.0175	235	5096.15	0.1931	420	9020.31	0.4526
55	1113.04	0.03	240	5204.82	0.0652	425	9119.1	0.1197
60	1222.34	0.0412	245	5313.28	0.0587	430	9212.93	0.1163
65	1331.46	0.0212	250	5421.83	0.066	435	9305.18	0.1425
70	1440.8	0.024	255	5530.16	0.0654	440	9397.67	0.1378
75	1550.75	0.0694	260	5638.34	0.0547	445	9492.1	0.1178
80	1686.96	0.0435	265	5746.46	0.0569	450	9586.14	0.1528
85	1797.99	0.0444	270	5854.22	0.0385	455	9680.63	0.1237
90	1908.45	0.0326	275	5962.24	0.061	460	9774.82	0.1517
95	2018.93	0.0399	280	6070.05	0.0664	465	9869.79	0.0913
100	2129.51	0.0401	285	6177.62	0.0961	470	9965.76	0.113
105	2240.12	0.0228	290	6285.12	0.0736	475	10062.86	0.138
110	2350.89	0.0525	295	6392.27	0.0381	480	10156.96	0.0836
115	2461.27	0.0418	300	6499.63	0.0748	485	10253.05	0.158
120	2571.7	0.0478	305	6611.6	0.1033	490	10350.12	0.1653
125	2682.45	0.0445	310	6715.57	0.1269	495	10445.48	0.0959
130	2792.72	0.0458	315	6823.79	0.0705	500	10539.8	0.1364
135	2903.06	0.1427	320	6929.17	0.0738	505	10636.4	0.1548
140	3013.42	0.0476	325	7035.12	0.1124	510	10731.04	0.1493
145	3123.85	0.0424	330	7141.31	0.1212	515	10826.27	0.1678
150	3233.92	0.0398	335	7246.4	0.0961	520	10921.43	0.1464
155	3344.12	0.052	340	7352.57	0.089	525	11016.23	0.1702
160	3454.27	0.0424	345	7456.92	0.1544	530	11108.44	0.1543
165	3564.33	0.0561	350	7561.77	0.0718	535	11195.23	0.053
170	3674.28	0.0496	355	7667.74	0.1285	540	11255.97	0.663
175	3784.42	0.046	360	7771.32	0.0715	545	11301.9	0.2122
180	3894.2	0.0484	365	7873.92	0.1297	550	11349.43	0.1452
185	4004.02	0.0641	370	7979.81	0.099	555	11396.14	0.0795

SetPnt: Rf Source Level Setting in mVpp

Filter: Average of 16 DMM measurements of the Filter Output from the Rectifier Module

Sigma: The Standard Deviation of the 16 DMM Filter measurements


SBCM Chain A Characteristic Curve
Slope: 21.44305mVa/mVrf
Intercept: 0.5575mVa
Correlation: 0.999787
Correlation>0.99?: PASS
Each point is the average of the 16 consecutive measurements taken at each RF Level setting. The fit is performed for RF settings between 5mVpp and 500mVpp
Data
Fit
Average Rectifier Filter Output (mV)
RF Level @ 499.68MHz (mV)
0
2000
4000
6000
8000
10000
0
100
200
300
400
500

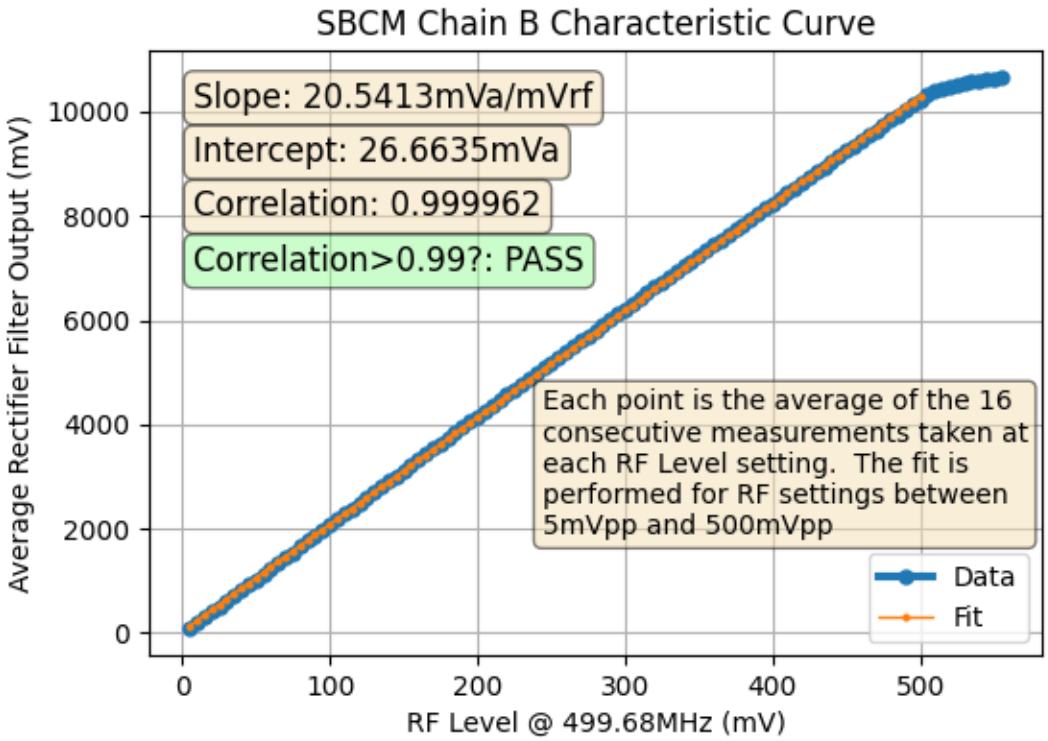

Each point is the standard deviation of the 16 measurements used to calculate the average Filter Output for each RF level setting shown above
max(Sigma)<4.0?: PASS
Sigma for Filter Output Average (mV)
RF Level @ 499.68MHz (mV)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
100
200
300
400
500

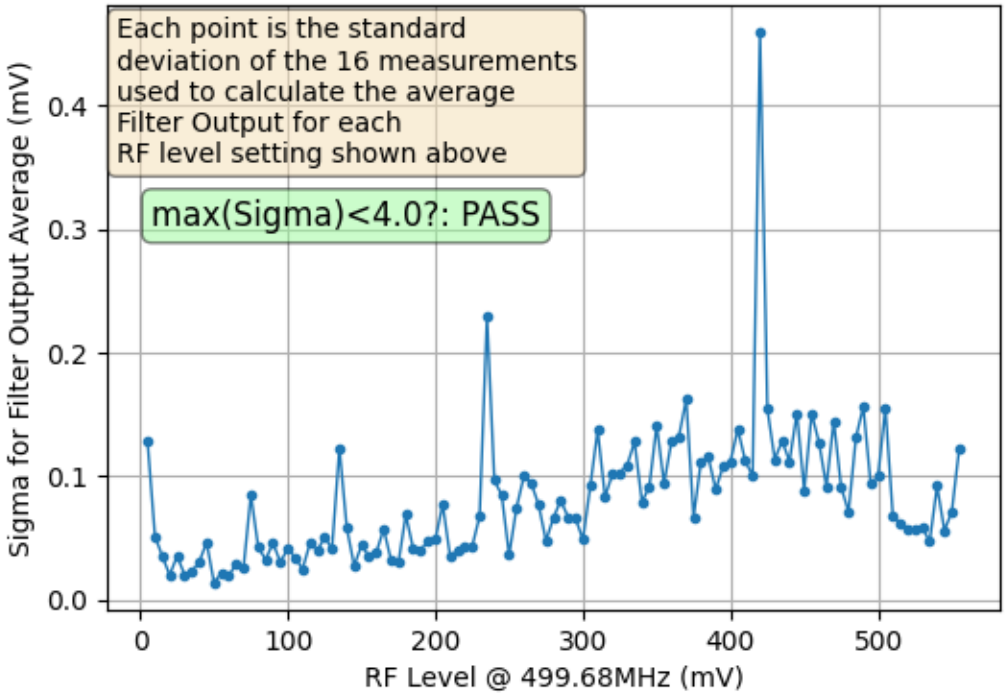
SBCM Chain-B Characteristic Curve				file: SBCM-B_2026_CharCurve.txt				
SetPnt (mV)	Filter (mV)	Sigma (mV)	SetPnt (mV)	Filter (mV)	Sigma (mV)	SetPnt (mV)	Filter (mV)	Sigma (mV)
5	98.24	0.1277	190	3949.18	0.0399	375	7741.0	0.0667
10	198.24	0.0514	195	4052.39	0.0473	380	7842.54	0.1119
15	299.82	0.035	200	4156.14	0.0488	385	7942.04	0.1153
20	401.85	0.0197	205	4259.81	0.0771	390	8042.1	0.0891
25	504.84	0.0353	210	4363.43	0.0349	395	8143.75	0.1079
30	607.62	0.0202	215	4466.88	0.04	400	8243.43	0.112
35	710.47	0.0228	220	4570.55	0.0425	405	8342.22	0.138
40	813.54	0.0303	225	4673.53	0.0431	410	8443.74	0.1125
45	918.1	0.0463	230	4777.14	0.0675	415	8543.04	0.1001
50	1021.39	0.0138	235	4879.76	0.2292	420	8648.93	0.4594
55	1124.49	0.0212	240	4982.9	0.0966	425	8748.81	0.1542
60	1227.94	0.0189	245	5085.94	0.0845	430	8848.45	0.1136
65	1331.17	0.0291	250	5189.17	0.036	435	8947.83	0.1285
70	1434.64	0.0253	255	5292.22	0.0743	440	9046.8	0.112
75	1538.84	0.085	260	5395.15	0.0998	445	9147.03	0.151
80	1660.04	0.0423	265	5498.07	0.0942	450	9245.78	0.0884
85	1764.65	0.0323	270	5600.77	0.0766	455	9344.68	0.1508
90	1868.67	0.0459	275	5703.71	0.0475	460	9443.01	0.1271
95	1972.75	0.03	280	5806.53	0.0662	465	9541.77	0.0916
100	2076.91	0.0415	285	5909.16	0.0803	470	9640.98	0.1446
105	2181.1	0.034	290	6011.84	0.0666	475	9741.42	0.0912
110	2285.48	0.0246	295	6114.28	0.0657	480	9838.59	0.0709
115	2389.47	0.0453	300	6216.98	0.0485	485	9937.52	0.1311
120	2493.54	0.0395	305	6324.29	0.0919	490	10037.47	0.1563
125	2597.94	0.0503	310	6423.82	0.1372	495	10135.33	0.0936
130	2701.83	0.0411	315	6527.58	0.0833	500	10231.1	0.0998
135	2805.81	0.1226	320	6628.5	0.1017	505	10319.96	0.1549
140	2909.93	0.0579	325	6730.18	0.1026	510	10389.31	0.0679
145	3014.1	0.0273	330	6832.24	0.1088	515	10442.82	0.0623
150	3117.94	0.044	335	6933.09	0.1279	520	10486.37	0.0574
155	3221.9	0.035	340	7035.21	0.079	525	10523.34	0.0564
160	3325.92	0.0381	345	7135.56	0.0913	530	10555.05	0.0583
165	3429.84	0.057	350	7236.63	0.1412	535	10582.9	0.0469
170	3533.73	0.0317	355	7338.83	0.0939	540	10606.6	0.0919
175	3637.76	0.0298	360	7438.54	0.1283	545	10625.49	0.0558
180	3741.49	0.0686	365	7537.73	0.1318	550	10640.16	0.0712
185	3845.37	0.0411	370	7639.78	0.1621	555	10650.21	0.1224

SetPnt: Rf Source Level Setting in mVpp

Filter: Average of 16 DMM measurements of the Filter Output from the Rectifier Module

Sigma: The Standard Deviation of the 16 DMM Filter measurements


SBCM Chain B Characteristic Curve
Slope: 20.5413mVa/mVrf
Intercept: 26.6635mVa
Correlation: 0.999962
Correlation>0.99?: PASS
Each point is the average of the 16 consecutive measurements taken at each RF Level setting. The fit is performed for RF settings between 5mVpp and 500mVpp
Data
Fit
Average Rectifier Filter Output (mV)
RF Level @ 499.68MHz (mV)
0
2000
4000
6000
8000
10000
0
100
200
300
400
500


Each point is the standard deviation of the 16 measurements used to calculate the average Filter Output for each RF level setting shown above
max(Sigma)<4.0?: PASS
Sigma for Filter Output Average (mV)
RF Level @ 499.68MHz (mV)
0.0
0.1
0.2
0.3
0.4
0
100
200
300
400
500

SBCM Beam Scan with 90% Fill

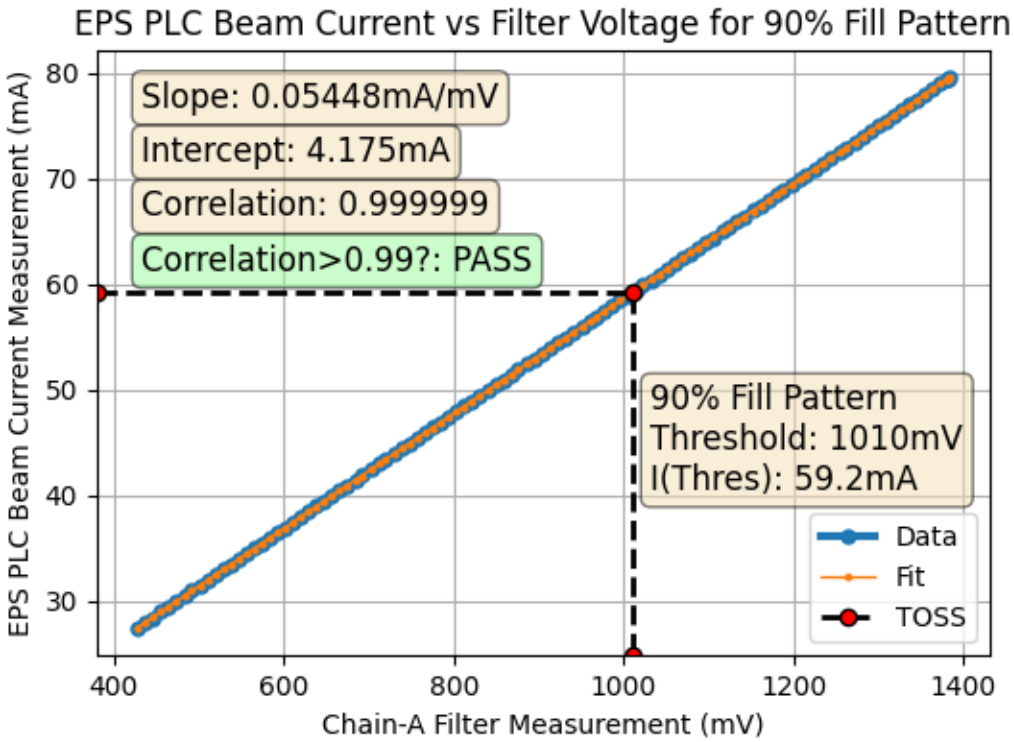
file: SBCM_BeamScanFull_90pctFill_2026.txt

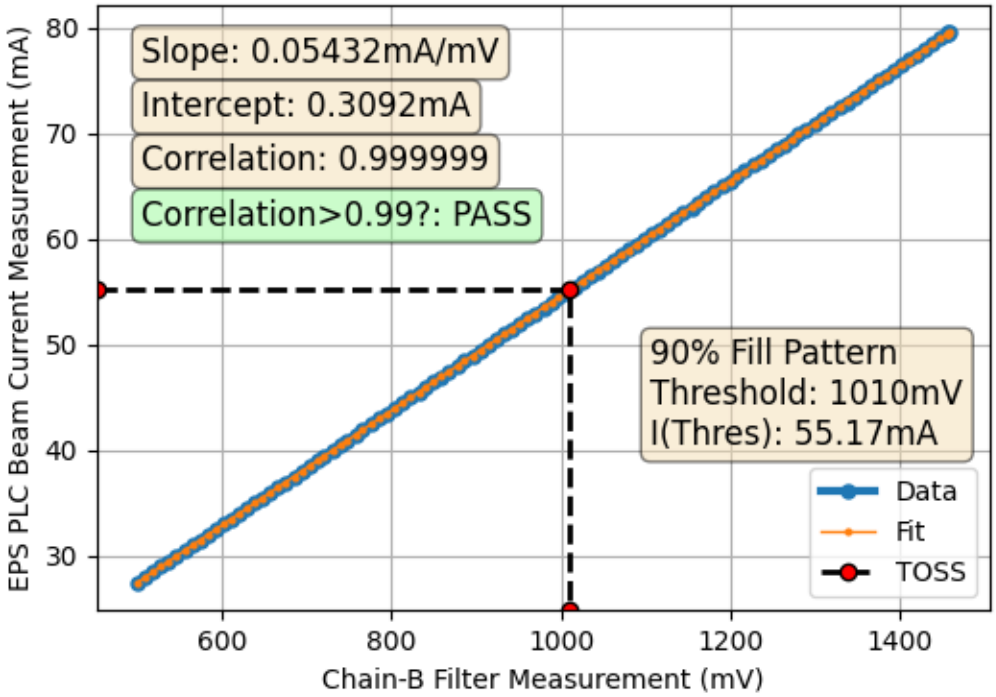
SR Beam PLC(mA)	Chain-A Filter(mV)	Chain-B Filter(mV)	SR Beam PLC(mA)	Chain-A Filter(mV)	Chain-B Filter(mV)	SR Beam PLC(mA)	Chain-A Filter(mV)	Chain-B Filter(mV)
79.53	1383.26	1458.28	62.95	1079.51	1153.75	44.98	748.7	822.49
79.5	1382.37	1457.42	62.49	1069.97	1144.5	44.44	739.46	813.15
79.47	1381.83	1456.92	61.92	1060.13	1134.7	43.98	730.94	804.11
79.4	1381.31	1456.42	61.46	1051.15	1125.64	43.41	719.94	793.57
78.97	1373.0	1448.07	60.89	1041.6	1115.85	42.98	712.72	786.35
78.47	1364.08	1439.09	60.42	1032.8	1107.34	42.48	702.58	775.69
77.94	1354.41	1429.38	59.96	1023.13	1097.67	41.91	692.16	765.66
77.47	1345.72	1420.76	59.49	1014.87	1089.39	41.48	684.48	757.49
76.97	1335.74	1410.78	58.96	1005.59	1079.21	40.91	674.39	747.83
76.47	1327.58	1402.56	58.46	995.74	1070.16	40.48	665.74	739.15
75.97	1317.82	1392.75	57.93	986.46	1060.92	39.95	656.45	729.74
75.44	1308.57	1383.6	57.46	977.95	1051.99	39.45	647.33	720.65
74.97	1300.01	1374.9	56.93	968.84	1043.02	38.98	638.33	711.63
74.47	1290.09	1365.02	56.5	960.2	1033.68	38.48	629.53	702.74
73.97	1281.53	1356.35	56.0	951.45	1025.45	37.98	620.86	693.95
73.48	1271.83	1346.51	55.43	941.13	1015.42	37.48	611.57	684.56
72.94	1262.44	1337.38	54.93	931.58	1005.09	36.95	601.32	674.4
72.48	1253.69	1328.53	54.43	922.65	996.92	36.49	593.2	666.18
71.91	1243.35	1318.23	54.0	914.5	988.7	35.99	583.21	655.98
71.44	1234.01	1308.82	53.5	905.41	979.6	35.49	574.64	647.42
70.94	1224.76	1299.58	52.9	894.73	968.81	34.99	565.2	638.04
70.38	1215.14	1289.54	52.47	886.63	960.74	34.45	555.83	628.67
69.95	1206.83	1280.47	51.9	876.01	950.08	33.99	547.17	619.83
69.48	1198.25	1272.9	51.44	867.6	941.44	33.49	537.79	610.58
68.95	1188.14	1262.92	50.97	858.6	932.65	32.99	529.35	602.03
68.48	1179.69	1254.44	50.47	849.67	923.79	32.49	519.56	592.16
67.92	1170.26	1244.58	49.97	840.42	913.75	31.96	510.46	582.94
67.48	1161.67	1236.37	49.44	830.83	904.64	31.49	501.49	574.01
66.98	1152.31	1227.0	48.94	821.47	895.24	30.99	492.12	564.41
66.42	1142.14	1216.79	48.47	812.0	885.89	30.49	483.65	555.56
65.89	1132.94	1207.56	47.94	803.63	877.54	29.96	473.26	545.53
65.45	1124.68	1199.3	47.44	793.98	867.27	29.43	463.86	536.09
64.99	1116.11	1190.29	46.94	785.14	858.76	29.0	455.8	527.91
64.42	1105.94	1180.51	46.44	775.24	849.09	28.43	445.86	517.81
63.95	1097.48	1171.74	45.98	766.64	840.42	28.0	437.83	509.42
63.45	1087.68	1162.24	45.48	758.15	831.9	27.5	428.23	499.92

SR Beam PLC: EPS PLC Measurement of Stored Beam Intensity

Filter ChainA: DMM measurement of Output from Chain-A Rectifier Module

Filter ChainB: DMM measurement of Output from Chain-B Rectifier Module


EPS PLC Beam Current vs Filter Voltage for 90% Fill Pattern
Slope: 0.05448mA/mV
Intercept: 4.175mA
Correlation: 0.999999
Correlation>0.99?: PASS
90% Fill Pattern
Threshold: 1010mV
I(Thres): 59.2mA
Data
Fit
TOSS
EPS PLC Beam Current Measurement (mA)
Chain-A Filter Measurement (mV)
80
70
60
50
40
30
400
600
800
1000
1200
1400


Slope: 0.05432mA/mV
Intercept: 0.3092mA
Correlation: 0.999999
Correlation>0.99?: PASS
90% Fill Pattern
Threshold: 1010mV
I(Thres): 55.17mA
Data
Fit
TOSS
EPS PLC Beam Current Measurement (mA)
Chain-B Filter Measurement (mV)
80
70
60
50
40
30
600
800
1000
1200
1400

SBCM Beam Scan with 20% Fill

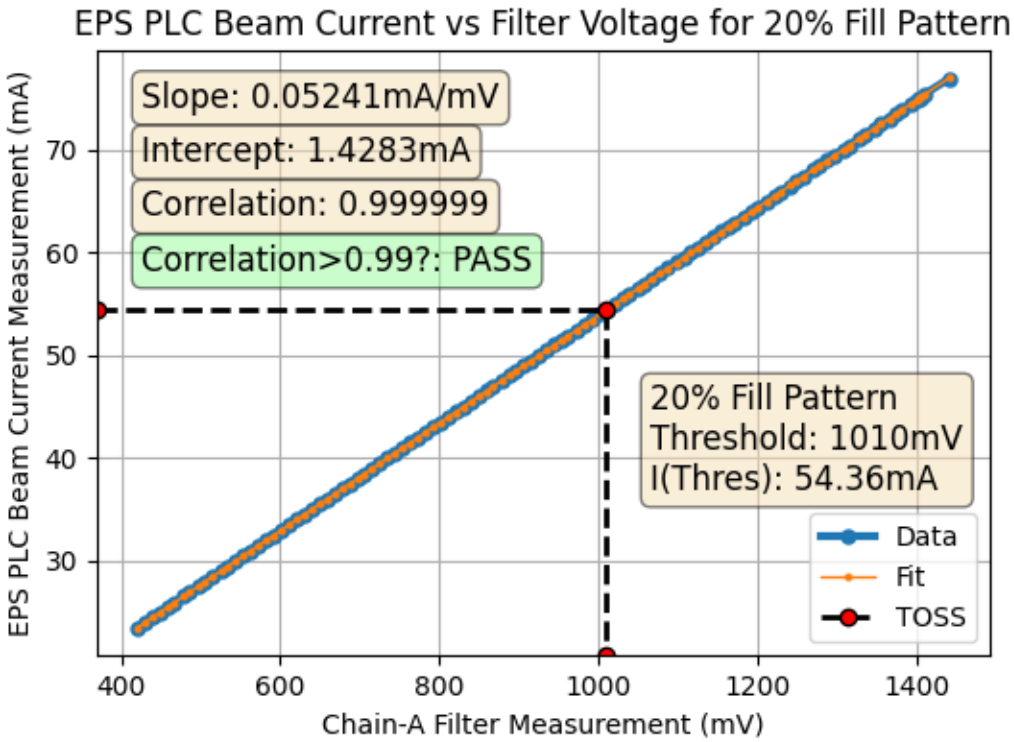
file: SBCM_BeamScanFull_20pctFill_2026.txt

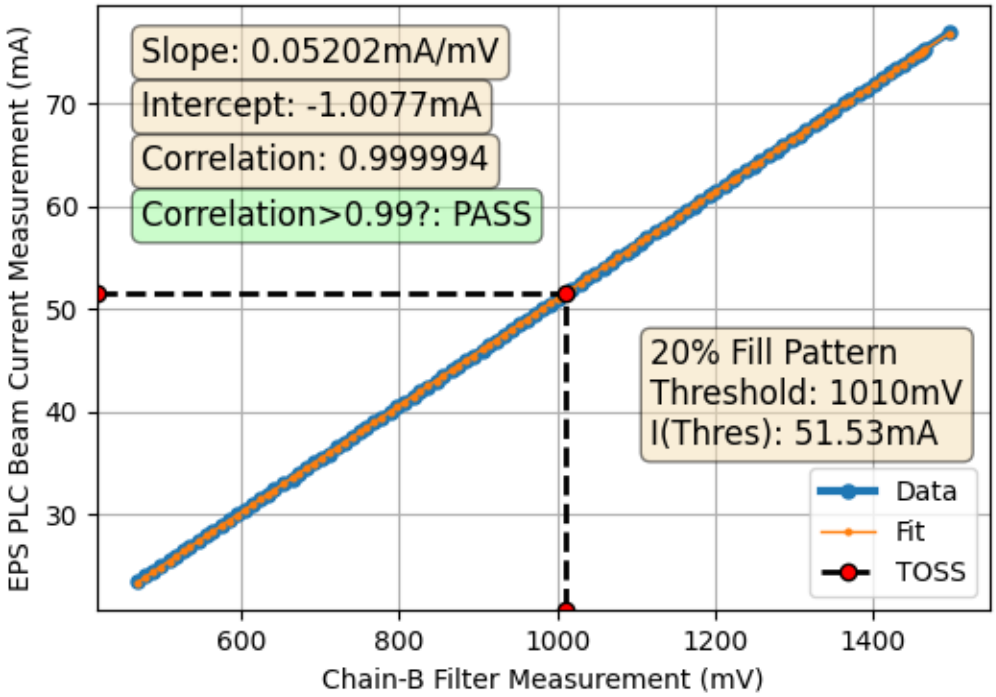
SR Beam PLC(mA)	Chain-A Filter(mV)	Chain-B Filter(mV)	SR Beam PLC(mA)	Chain-A Filter(mV)	Chain-B Filter(mV)	SR Beam PLC(mA)	Chain-A Filter(mV)	Chain-B Filter(mV)
76.87	1440.84	1495.94	58.99	1097.78	1154.49	40.91	753.96	806.65
75.31	1408.61	1464.57	58.46	1087.32	1143.71	40.45	744.67	797.02
75.04	1405.46	1461.15	57.93	1077.87	1134.09	39.98	735.05	788.57
74.87	1402.29	1458.63	57.43	1068.14	1124.21	39.42	724.71	777.6
74.71	1399.15	1454.57	56.86	1056.74	1112.97	38.95	715.74	768.41
74.34	1392.35	1447.2	56.43	1049.17	1105.51	38.42	705.89	758.49
73.84	1381.96	1437.99	55.9	1039.19	1094.71	37.98	697.15	750.29
73.44	1374.13	1429.02	55.46	1030.93	1086.96	37.48	687.79	739.5
72.94	1365.19	1420.99	55.0	1021.07	1076.87	36.95	677.83	729.95
72.44	1355.41	1411.4	54.43	1011.48	1067.83	36.49	668.96	720.34
71.98	1346.02	1401.18	53.97	1002.05	1058.5	35.92	658.53	710.46
71.41	1335.72	1391.81	53.4	991.18	1047.26	35.49	650.09	702.18
70.94	1327.03	1382.84	52.93	982.63	1038.08	34.99	639.93	691.8
70.35	1314.61	1368.94	52.43	973.17	1028.51	34.45	630.33	682.68
69.95	1307.47	1363.81	51.87	962.1	1017.42	33.99	621.08	672.24
69.35	1296.28	1351.45	51.44	954.14	1009.1	33.49	611.96	664.4
68.91	1286.97	1343.16	50.97	944.92	1000.8	32.96	601.99	653.5
68.38	1277.44	1334.02	50.47	935.23	990.62	32.46	591.17	641.23
67.95	1269.31	1325.58	49.97	925.6	980.77	31.93	581.73	632.35
67.38	1258.32	1314.74	49.44	915.97	971.05	31.49	573.37	623.93
66.88	1249.08	1305.61	48.94	906.45	960.93	30.93	562.93	613.95
66.45	1240.95	1297.19	48.47	896.92	951.87	30.46	554.1	604.09
65.89	1229.8	1285.77	47.97	888.15	942.68	29.99	544.38	594.77
65.42	1221.03	1277.67	47.47	877.8	932.36	29.39	533.5	584.01
64.92	1211.18	1266.9	46.87	867.25	921.68	28.96	525.59	574.79
64.39	1200.59	1255.72	46.47	859.16	913.53	28.4	514.86	563.54
63.92	1192.13	1249.25	45.98	849.39	905.05	27.96	505.75	555.75
63.39	1182.21	1238.38	45.44	839.33	893.11	27.5	498.09	546.88
62.85	1171.17	1226.94	44.98	830.66	884.85	26.86	485.64	534.87
62.45	1164.37	1220.81	44.44	820.2	873.83	26.5	479.04	527.74
61.99	1154.89	1210.88	43.98	812.06	866.93	25.9	467.31	516.08
61.42	1144.32	1200.75	43.41	800.7	853.87	25.5	459.92	508.49
60.96	1135.59	1192.28	42.98	792.69	846.35	24.93	449.16	497.73
60.46	1126.12	1181.38	42.41	781.8	835.67	24.47	440.64	488.89
59.96	1116.31	1173.84	41.91	772.46	825.94	23.97	431.14	478.89
59.49	1107.49	1163.49	41.45	763.1	817.12	23.47	420.75	468.62

SR Beam PLC: EPS PLC Measurement of Stored Beam Intensity

Filter ChainA: DMM measurement of Output from Chain-A Rectifier Module

Filter ChainB: DMM measurement of Output from Chain-B Rectifier Module





Based on the data shown above the following changes were made:
I(TOSS) = 54mA for a 20% Fill Pattern
Chain-A Threshold set to 1010mV
Chain-B Threshold set to 1010mV

SBCM Functional Scan with 20% Fill and Increasing Beam

file: SBCM_BeamScanFull_20pctFill_2026.txt

Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R	Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R
0.03	8.57	0,0,0	34.9	0,0,0	35.16	644.83	1,0,0	698.33	1,0,0
0.46	11.24	1,0,0	41.92	1,0,0	36.05	662.03	1,0,0	715.58	1,0,0
0.46	11.25	1,0,0	41.89	1,0,0	37.09	681.56	1,0,0	736.19	1,0,0
1.03	19.97	1,0,0	52.86	1,0,0	38.15	702.15	1,0,0	757.02	1,0,0
2.26	35.92	1,0,0	71.77	1,0,0	39.02	719.25	1,0,0	774.38	1,0,0
3.19	49.27	1,0,0	87.37	1,0,0	40.15	740.2	1,0,0	794.83	1,0,0
4.16	66.06	1,0,0	104.44	1,0,0	41.01	756.47	1,0,0	811.26	1,0,0
5.02	81.1	1,0,0	119.75	1,0,0	42.01	775.15	1,0,0	831.23	1,0,0
6.02	97.62	1,0,0	138.64	1,0,0	43.15	797.7	1,0,0	853.91	1,0,0
7.05	116.48	1,0,0	158.26	1,0,0	44.14	816.58	1,0,0	872.73	1,0,0
8.02	134.69	1,0,0	176.48	1,0,0	45.01	832.98	1,0,0	888.72	1,0,0
9.15	155.8	1,0,0	197.54	1,0,0	46.08	853.6	1,0,0	910.59	1,0,0
10.05	172.07	1,0,0	214.36	1,0,0	47.01	871.23	1,0,0	928.59	1,0,0
11.15	191.74	1,0,0	236.35	1,0,0	48.11	892.29	1,0,0	949.73	1,0,0
12.12	209.78	1,0,0	254.24	1,0,0	49.14	911.96	1,0,0	969.08	1,0,0
13.08	227.94	1,0,0	272.41	1,0,0	50.04	929.14	1,0,0	987.02	1,0,0
14.05	246.38	1,0,0	290.85	1,0,0	51.07	949.02	1,0,0	1007.16	1,0,0
15.04	264.44	1,0,0	310.22	1,0,0	52.07	968.26	1,0,0	1026.62	1,0,0
16.01	282.27	1,0,0	329.05	1,0,0	53.13	988.6	1,0,0	1046.41	1,0,0
17.14	303.78	1,0,0	350.37	1,0,0	53.53	995.95	1,0,0	1053.54	1,2,3
18.14	322.57	1,0,0	369.16	1,0,0	53.53	995.95	1,0,0	1053.55	1,2,3
19.14	341.21	1,0,0	388.49	1,0,0	53.73	999.41	1,2,0	1057.47	1,2,3
20.07	358.29	1,0,0	406.9	1,0,0	53.73	999.37	1,2,3	1057.51	1,2,3
21.0	376.25	1,0,0	424.78	1,0,0	54.06	1005.45	1,2,3	1063.84	1,2,3
22.07	396.47	1,0,0	445.15	1,0,0	55.13	1025.46	1,2,3	1084.97	1,2,3
23.0	413.84	1,0,0	462.57	1,0,0	56.16	1045.35	1,2,3	1104.84	1,2,3
24.0	433.32	1,0,0	483.58	1,0,0	57.03	1062.54	1,2,3	1121.45	1,2,3
25.03	452.08	1,0,0	502.66	1,0,0	58.06	1081.57	1,2,3	1140.51	1,2,3
26.03	474.15	1,0,0	524.63	1,0,0	59.16	1103.3	1,2,3	1163.44	1,2,3
27.1	491.7	1,0,0	542.13	1,0,0	60.12	1121.35	1,2,3	1181.57	1,2,3
28.06	509.61	1,0,0	561.07	1,0,0	61.16	1141.43	1,2,3	1201.43	1,2,3
29.03	528.26	1,0,0	580.34	1,0,0	62.06	1158.51	1,2,3	1217.9	1,2,3
30.09	549.17	1,0,0	601.43	1,0,0	63.12	1178.83	1,2,3	1239.34	1,2,3
31.03	567.2	1,0,0	619.22	1,0,0	64.15	1198.47	1,2,3	1259.6	1,2,3
32.02	585.26	1,0,0	637.59	1,0,0	65.05	1216.18	1,2,3	1277.5	1,2,3
33.06	605.32	1,0,0	658.97	1,0,0	66.08	1236.15	1,2,3	1296.74	1,2,3
34.09	624.59	1,0,0	678.26	1,0,0	67.08	1254.58	1,2,3	1315.03	1,2,3

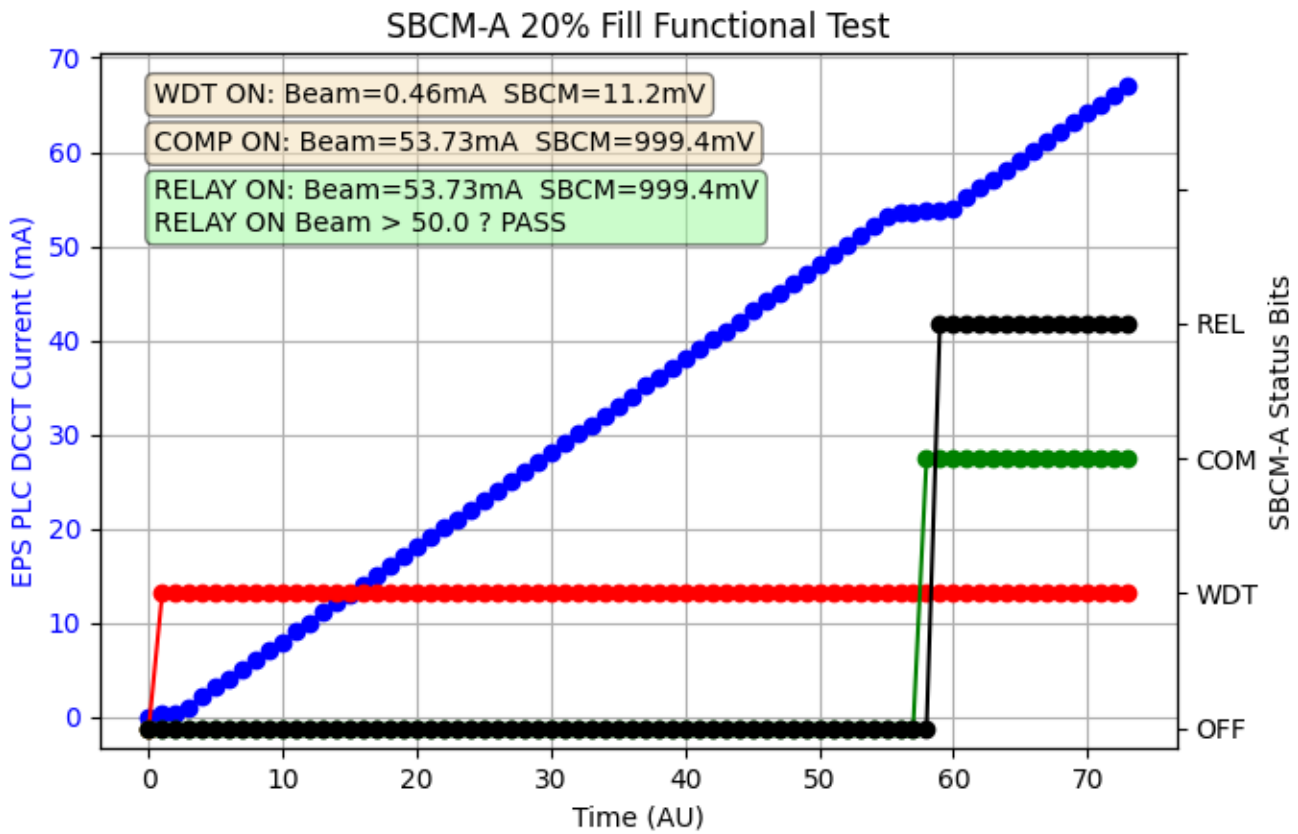
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

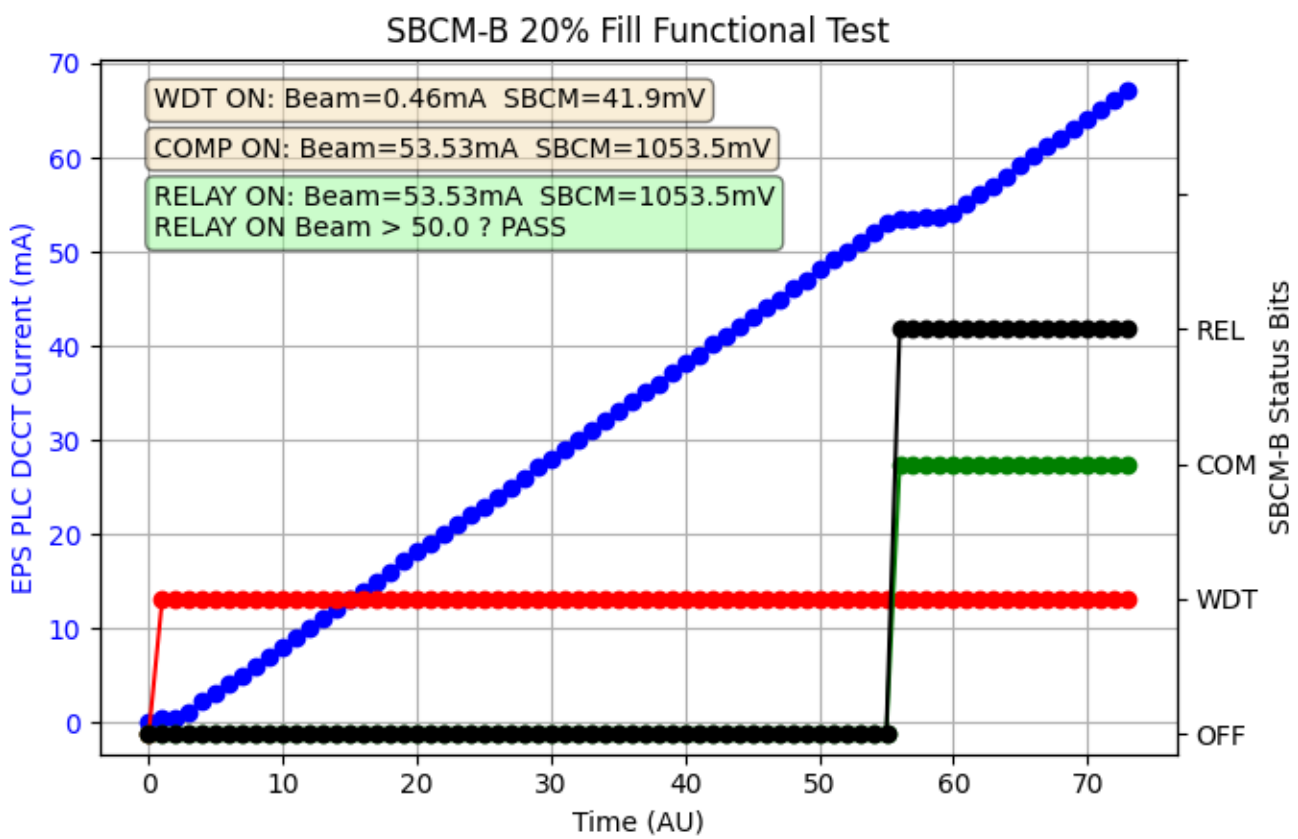
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC


SBCM-A 20% Fill Functional Test
WDT ON: Beam=0.46mA SBCM=11.2mV
COMP ON: Beam=53.73mA SBCM=999.4mV
RELAY ON: Beam=53.73mA SBCM=999.4mV
RELAY ON Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-A Status Bits
REL
COM
WDT
OFF
Time (AU)


SBCM-B 20% Fill Functional Test
WDT ON: Beam=0.46mA SBCM=41.9mV
COMP ON: Beam=53.53mA SBCM=1053.5mV
RELAY ON: Beam=53.53mA SBCM=1053.5mV
RELAY ON Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-B Status Bits
REL
COM
WDT
OFF
Time (AU)

SBCM Functional Scan with 20% Fill and Decreasing Beam

file: SBCM_FunctionalTestFalling_20pctFill_2026.txt

Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R	Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R
59.99	1117.62	1,2,3	1173.15	1,2,3	30.99	563.05	1,0,0	613.78	1,0,0
58.99	1098.2	1,2,3	1154.01	1,2,3	29.99	544.36	1,0,0	594.65	1,0,0
57.96	1078.25	1,2,3	1133.98	1,2,3	29.0	526.0	1,0,0	576.01	1,0,0
56.93	1058.9	1,2,3	1114.84	1,2,3	28.0	506.63	1,0,0	556.25	1,0,0
56.0	1040.19	1,2,3	1096.11	1,2,3	26.96	487.48	1,0,0	536.78	1,0,0
55.0	1021.99	1,2,3	1077.74	1,2,3	25.93	468.14	1,0,0	517.16	1,0,0
54.0	1002.67	1,2,3	1058.49	1,2,3	24.93	448.11	1,0,0	496.73	1,0,0
53.5	993.49	1,2,3	1049.16	1,0,0	23.9	429.05	1,0,0	477.33	1,0,0
53.5	993.49	1,2,3	1049.16	1,0,0	22.9	410.02	1,0,0	457.94	1,0,0
52.97	983.22	1,2,3	1038.55	1,0,0	21.94	391.33	1,0,0	438.97	1,0,0
52.67	978.12	1,0,3	1032.6	1,0,0	20.97	373.26	1,0,0	420.54	1,0,0
52.63	976.27	1,0,0	1031.32	1,0,0	19.97	354.61	1,0,0	401.52	1,0,0
51.9	962.67	1,0,0	1017.82	1,0,0	18.94	335.59	1,0,0	382.0	1,0,0
50.9	944.29	1,0,0	998.75	1,0,0	17.94	317.06	1,0,0	363.02	1,0,0
49.94	925.35	1,0,0	979.55	1,0,0	16.98	299.81	1,0,0	345.53	1,0,0
48.94	905.92	1,0,0	960.58	1,0,0	15.98	279.96	1,0,0	325.11	1,0,0
47.97	887.81	1,0,0	942.24	1,0,0	14.95	261.4	1,0,0	305.99	1,0,0
46.94	867.75	1,0,0	922.03	1,0,0	13.98	243.66	1,0,0	287.87	1,0,0
45.91	848.2	1,0,0	902.27	1,0,0	12.98	224.91	1,0,0	268.51	1,0,0
44.98	829.81	1,0,0	883.62	1,0,0	11.95	206.31	1,0,0	249.48	1,0,0
43.94	810.92	1,0,0	864.66	1,0,0	10.98	187.97	1,0,0	230.58	1,0,0
42.98	792.29	1,0,0	845.72	1,0,0	9.98	169.37	1,0,0	211.36	1,0,0
41.95	772.46	1,0,0	825.89	1,0,0	8.95	151.27	1,0,0	192.66	1,0,0
40.98	754.18	1,0,0	807.32	1,0,0	7.99	133.63	1,0,0	174.22	1,0,0
39.98	735.82	1,0,0	788.37	1,0,0	6.99	115.85	1,0,0	155.86	1,0,0
38.98	715.98	1,0,0	768.66	1,0,0	5.99	97.64	1,0,0	137.01	1,0,0
37.95	696.41	1,0,0	748.72	1,0,0	4.99	79.81	1,0,0	118.42	1,0,0
36.95	677.47	1,0,0	729.72	1,0,0	3.99	62.47	1,0,0	100.22	1,0,0
35.99	658.42	1,0,0	709.88	1,0,0	2.99	45.1	1,0,0	81.81	1,0,0
34.95	639.32	1,0,0	690.95	1,0,0	1.99	28.05	1,0,0	63.64	1,0,0
33.99	620.85	1,0,0	672.27	1,0,0	0.99	11.66	1,0,0	46.03	1,0,0
32.99	601.93	1,0,0	653.03	1,0,0	0.83	9.86	0,0,0	43.84	0,0,0
31.93	582.29	1,0,0	633.18	1,0,0	0.83	9.86	0,0,0	43.84	0,0,0

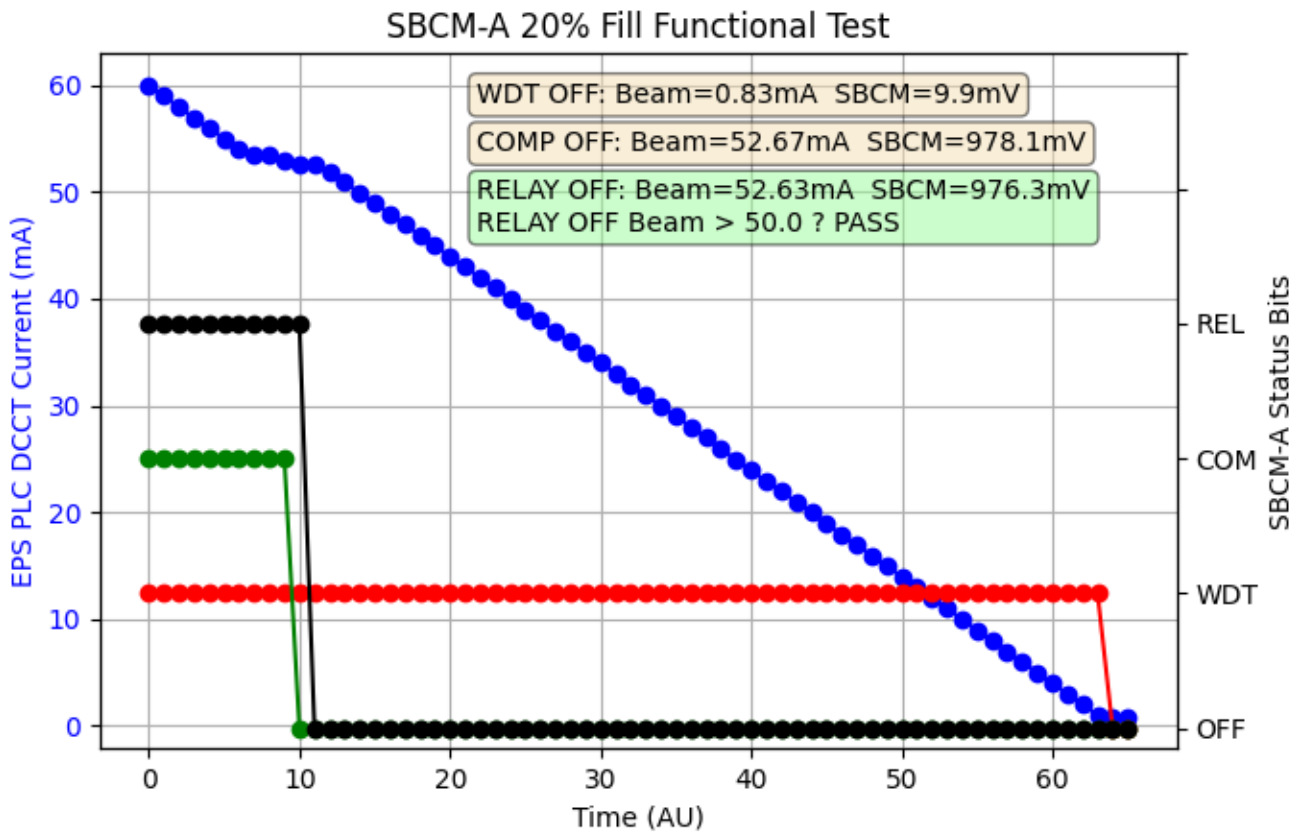
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

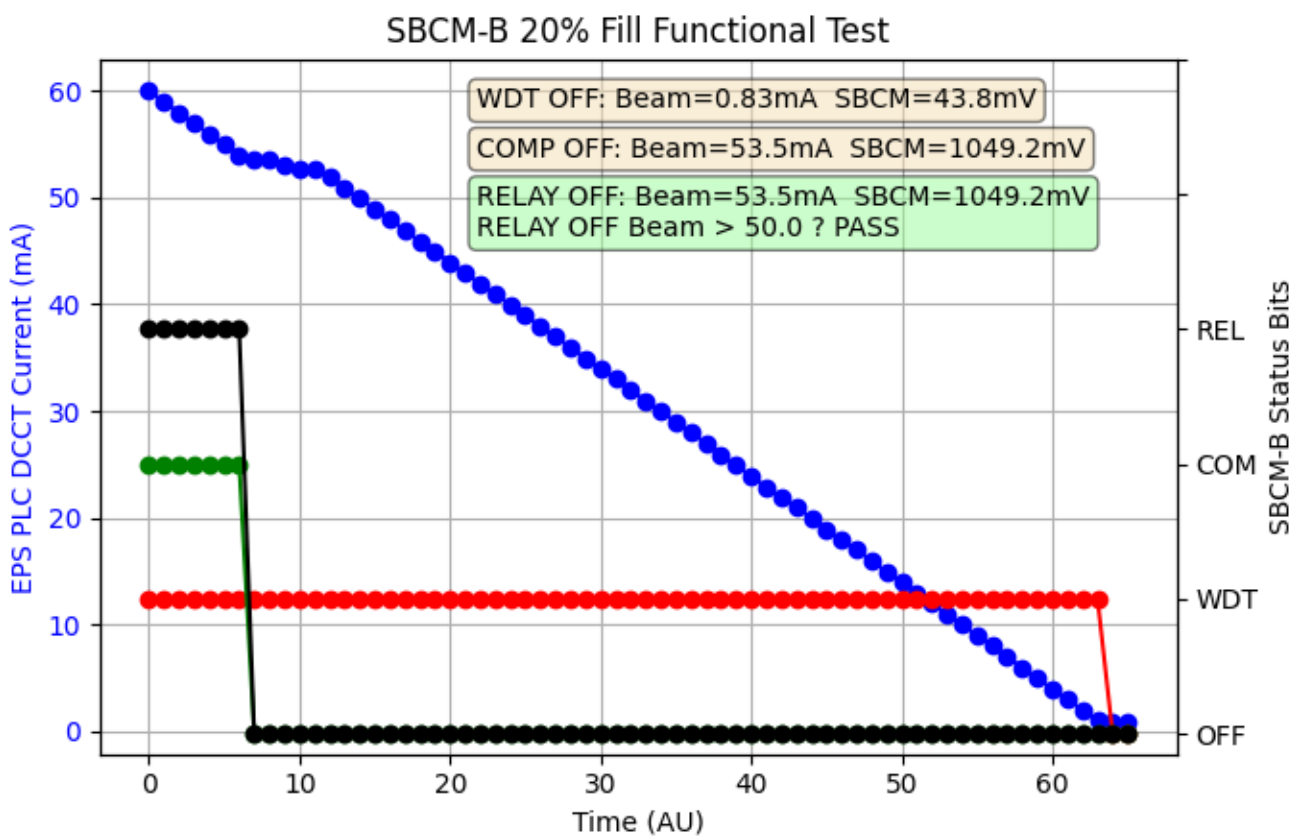
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC


SBCM-A 20% Fill Functional Test
WDT OFF: Beam=0.83mA SBCM=9.9mV
COMP OFF: Beam=52.67mA SBCM=978.1mV
RELAY OFF: Beam=52.63mA SBCM=976.3mV
RELAY OFF Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-A Status Bits
REL
COM
WDT
OFF
Time (AU)


SBCM-B 20% Fill Functional Test
WDT OFF: Beam=0.83mA SBCM=43.8mV
COMP OFF: Beam=53.5mA SBCM=1049.2mV
RELAY OFF: Beam=53.5mA SBCM=1049.2mV
RELAY OFF Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-B Status Bits
REL
COM
WDT
OFF
Time (AU)

SBCM Functional Scan with 90% Fill and Increasing Beam

file: SBCM_FunctionalTestRising_90pctFill_2026.txt

Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R	Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R
0.03	7.64	0,0,0	34.98	0,0,0	35.16	568.53	1,0,0	635.71	1,0,0
0.53	11.98	1,0,0	43.47	1,0,0	36.09	585.9	1,0,0	652.92	1,0,0
0.53	11.98	1,0,0	43.47	1,0,0	37.02	602.88	1,0,0	669.59	1,0,0
1.03	19.63	1,0,0	52.4	1,0,0	38.15	624.02	1,0,0	690.52	1,0,0
2.06	36.73	1,0,0	72.25	1,0,0	39.18	643.38	1,0,0	709.72	1,0,0
3.36	58.26	1,0,0	94.99	1,0,0	40.22	662.15	1,0,0	728.48	1,0,0
4.03	69.47	1,0,0	106.66	1,0,0	41.21	680.29	1,0,0	746.57	1,0,0
5.16	87.22	1,0,0	125.13	1,0,0	42.08	696.13	1,0,0	762.3	1,0,0
6.02	100.57	1,0,0	139.47	1,0,0	43.18	716.2	1,0,0	782.12	1,0,0
7.12	116.49	1,0,0	156.97	1,0,0	44.14	733.7	1,0,0	799.37	1,0,0
8.19	132.92	1,0,0	175.32	1,0,0	45.18	753.05	1,0,0	818.5	1,0,0
9.05	145.97	1,0,0	190.09	1,0,0	46.24	773.11	1,0,0	838.47	1,0,0
10.05	162.9	1,0,0	208.05	1,0,0	47.21	790.52	1,0,0	855.82	1,0,0
11.05	179.16	1,0,0	224.74	1,0,0	48.27	809.65	1,0,0	874.94	1,0,0
12.12	196.08	1,0,0	241.96	1,0,0	49.2	827.49	1,0,0	892.62	1,0,0
13.11	211.55	1,0,0	258.54	1,0,0	50.14	843.62	1,0,0	908.62	1,0,0
14.05	225.19	1,0,0	273.64	1,0,0	51.23	864.25	1,0,0	928.89	1,0,0
15.08	240.5	1,0,0	291.16	1,0,0	52.13	881.44	1,0,0	945.98	1,0,0
16.21	256.88	1,0,0	309.58	1,0,0	53.07	898.29	1,0,0	962.91	1,0,0
17.24	274.63	1,0,0	328.15	1,0,0	54.1	917.05	1,0,0	981.78	1,0,0
18.01	287.28	1,0,0	341.06	1,0,0	55.13	936.26	1,0,0	1001.34	1,0,0
19.11	305.05	1,0,0	358.99	1,0,0	56.1	954.3	1,0,0	1019.46	1,0,0
20.04	319.56	1,0,0	374.79	1,0,0	57.09	971.55	1,0,0	1037.22	1,0,0
21.2	336.07	1,0,0	393.48	1,0,0	58.13	989.34	1,0,0	1056.03	1,0,0
22.04	347.41	1,0,0	406.68	1,0,0	58.33	992.74	1,0,0	1059.49	1,2,3
23.07	363.21	1,0,0	424.57	1,0,0	58.33	992.74	1,0,0	1059.49	1,2,3
24.0	379.9	1,0,0	441.78	1,0,0	58.86	1001.82	1,2,3	1068.6	1,2,3
25.17	399.21	1,0,0	461.16	1,0,0	58.86	1001.82	1,2,3	1068.6	1,2,3
26.03	413.56	1,0,0	475.69	1,0,0	59.16	1008.08	1,2,3	1074.83	1,2,3
27.16	429.54	1,0,0	493.43	1,0,0	60.16	1025.85	1,2,3	1092.59	1,2,3
28.13	442.37	1,0,0	508.32	1,0,0	61.06	1043.51	1,2,3	1110.24	1,2,3
29.09	456.7	1,0,0	524.02	1,0,0	62.12	1062.93	1,2,3	1129.59	1,2,3
30.09	476.21	1,0,0	543.58	1,0,0	63.09	1080.48	1,2,3	1146.92	1,2,3
31.23	496.75	1,0,0	564.02	1,0,0	64.05	1097.55	1,2,3	1163.86	1,2,3
32.06	511.71	1,0,0	579.03	1,0,0	65.15	1118.06	1,2,3	1184.27	1,2,3
33.12	531.61	1,0,0	598.8	1,0,0	66.15	1135.85	1,2,3	1201.91	1,2,3
34.02	548.57	1,0,0	615.7	1,0,0	67.02	1151.97	1,2,3	1217.75	1,2,3

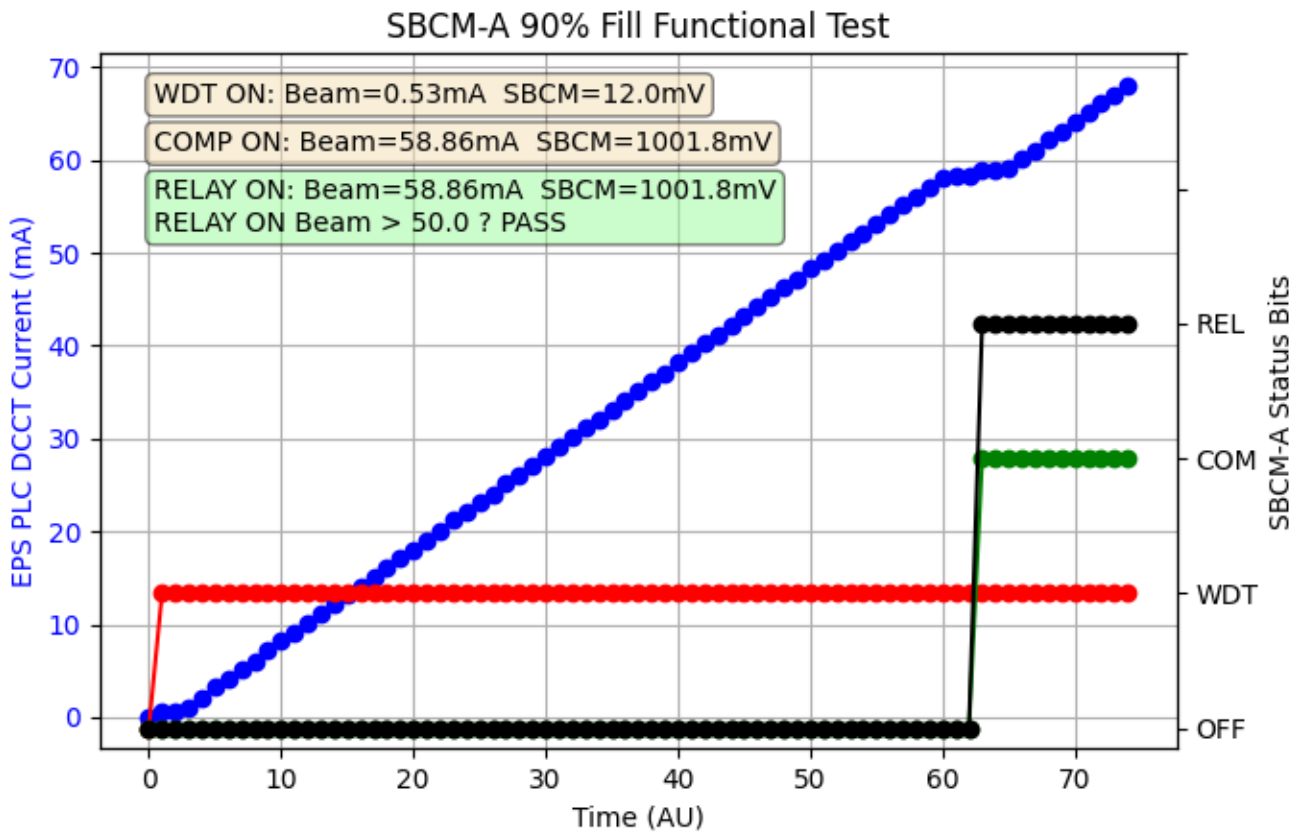
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

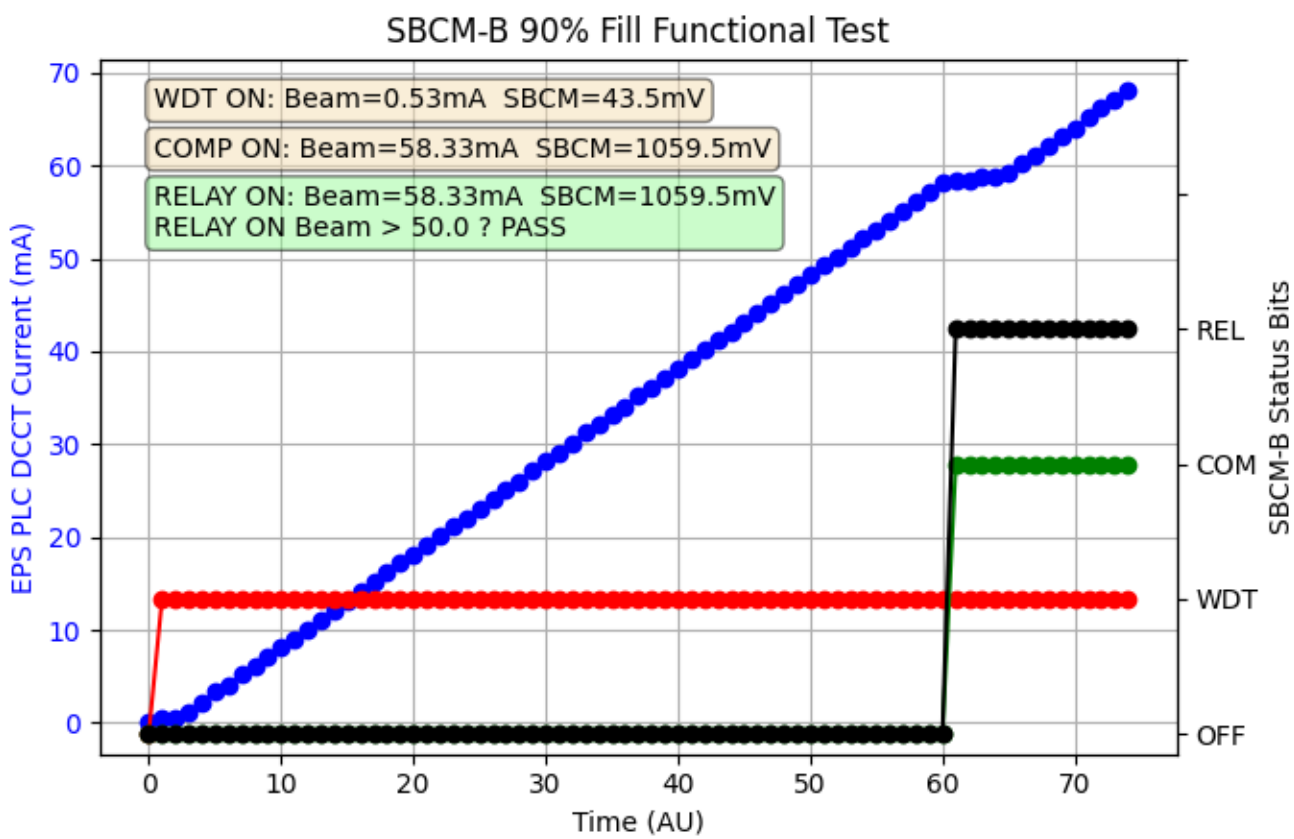
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC


SBCM-A 90% Fill Functional Test
WDT ON: Beam=0.53mA SBCM=12.0mV
COMP ON: Beam=58.86mA SBCM=1001.8mV
RELAY ON: Beam=58.86mA SBCM=1001.8mV
RELAY ON Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-A Status Bits
REL
COM
WDT
OFF
Time (AU)


SBCM-B 90% Fill Functional Test
WDT ON: Beam=0.53mA SBCM=43.5mV
COMP ON: Beam=58.33mA SBCM=1059.5mV
RELAY ON: Beam=58.33mA SBCM=1059.5mV
RELAY ON Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-B Status Bits
REL
COM
WDT
OFF
Time (AU)

SBCM Functional Scan with 90% Fill and Decreasing Beam

file: SBCM_FunctionalTestFalling_90pctFill_2026.txt

Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R	Beam (mA)	Filt-A (mV)	Bits-A W,C,R	Filt-B (mV)	Bits-B W,C,R
69.98	1207.46	1,2,3	1271.92	1,2,3	36.92	601.08	1,0,0	667.81	1,0,0
68.98	1188.72	1,2,3	1253.51	1,2,3	35.99	583.49	1,0,0	650.25	1,0,0
67.95	1170.16	1,2,3	1235.01	1,2,3	34.95	564.72	1,0,0	631.46	1,0,0
66.98	1151.89	1,2,3	1216.85	1,2,3	33.99	547.52	1,0,0	614.31	1,0,0
65.98	1133.67	1,2,3	1198.71	1,2,3	32.99	529.21	1,0,0	595.97	1,0,0
64.99	1115.81	1,2,3	1180.94	1,2,3	31.89	509.05	1,0,0	575.42	1,0,0
63.99	1097.79	1,2,3	1163.0	1,2,3	30.79	488.61	1,0,0	554.9	1,0,0
62.95	1078.35	1,2,3	1143.66	1,2,3	29.73	468.42	1,0,0	534.71	1,0,0
61.99	1060.38	1,2,3	1125.73	1,2,3	28.7	449.0	1,0,0	515.23	1,0,0
60.99	1042.64	1,2,3	1108.01	1,2,3	27.7	430.0	1,0,0	496.18	1,0,0
59.99	1024.42	1,2,3	1089.92	1,2,3	26.93	417.51	1,0,0	483.85	1,0,0
58.99	1006.01	1,2,3	1071.6	1,2,3	25.93	400.28	1,0,0	466.55	1,0,0
57.96	986.99	1,2,3	1052.55	1,2,3	24.97	382.71	1,0,0	448.9	1,0,0
57.66	980.86	1,2,3	1046.54	1,0,0	23.97	364.36	1,0,0	429.61	1,0,0
57.66	980.86	1,2,3	1046.54	1,0,0	22.94	345.98	1,0,0	412.0	1,0,0
57.49	978.37	1,2,0	1044.1	1,0,0	21.87	326.45	1,0,0	392.36	1,0,0
57.49	978.0	1,0,0	1043.71	1,0,0	20.84	307.67	1,0,0	373.45	1,0,0
56.99	968.89	1,0,0	1034.65	1,0,0	19.87	289.86	1,0,0	355.5	1,0,0
56.0	951.01	1,0,0	1016.85	1,0,0	18.97	273.28	1,0,0	338.76	1,0,0
55.0	932.85	1,0,0	998.82	1,0,0	17.94	255.48	1,0,0	320.74	1,0,0
54.0	914.74	1,0,0	980.62	1,0,0	16.98	238.39	1,0,0	303.49	1,0,0
52.97	895.63	1,0,0	961.77	1,0,0	15.91	219.22	1,0,0	284.07	1,0,0
51.9	876.24	1,0,0	942.48	1,0,0	14.95	201.43	1,0,0	266.04	1,0,0
50.87	857.33	1,0,0	923.6	1,0,0	13.98	184.59	1,0,0	248.76	1,0,0
49.97	840.18	1,0,0	906.5	1,0,0	12.98	166.33	1,0,0	230.3	1,0,0
48.97	821.94	1,0,0	888.3	1,0,0	11.98	149.27	1,0,0	212.89	1,0,0
47.97	803.25	1,0,0	869.63	1,0,0	10.92	130.49	1,0,0	193.61	1,0,0
46.94	785.14	1,0,0	851.6	1,0,0	9.95	113.64	1,0,0	176.28	1,0,0
45.91	765.91	1,0,0	832.42	1,0,0	8.95	95.52	1,0,0	157.42	1,0,0
44.98	747.96	1,0,0	814.48	1,0,0	7.89	78.37	1,0,0	139.55	1,0,0
43.98	730.2	1,0,0	796.79	1,0,0	6.92	61.96	1,0,0	122.19	1,0,0
42.98	712.3	1,0,0	778.73	1,0,0	5.99	45.89	1,0,0	104.92	1,0,0
41.95	693.09	1,0,0	759.75	1,0,0	4.89	30.25	1,0,0	87.25	1,0,0
40.91	674.5	1,0,0	741.21	1,0,0	3.96	17.91	1,0,0	70.93	1,0,0
39.98	656.47	1,0,0	723.18	1,0,0	3.66	15.11	1,0,0	66.07	0,0,0
38.95	638.3	1,0,0	705.0	1,0,0	3.33	12.62	0,0,0	60.5	0,0,0
37.98	620.81	1,0,0	687.52	1,0,0	2.99	10.86	0,0,0	55.09	0,0,0

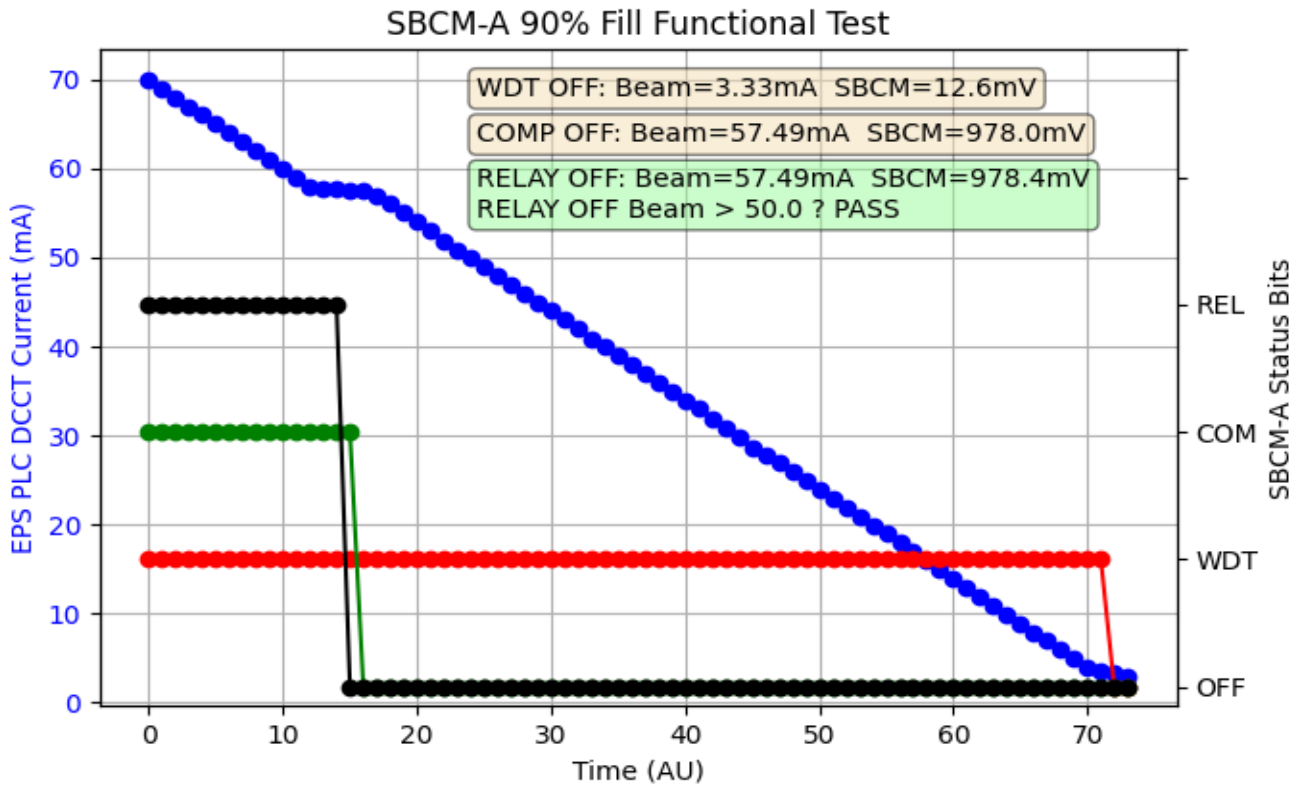
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

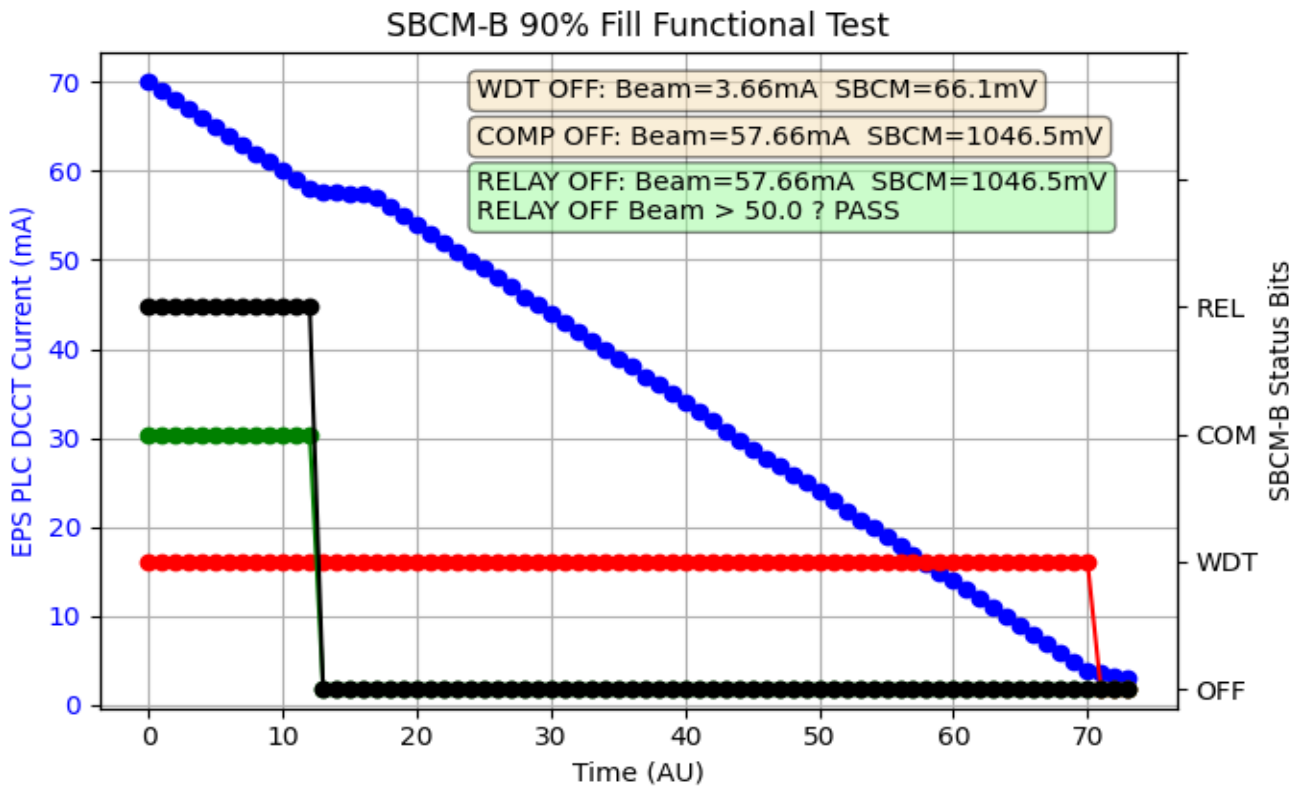
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC





All Tests for this Procedure have PASSED
TOSS SBCM Calibration Final Result: PASS